SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:20**

Annexure A

CLASS TEST

Criteria	Understanding of Task *(2)*	Procedure & Execution *(2.5)*	Observation & Result *(2)*	Viva Voce / Conceptual Response *(2)*	Time Management & Presentation *(1.5)*	***Total(10)***
Weightage	*20%*	25%	20%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

I A. LOKESH KUMAR (192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS: **Total Marks: 10x2 =20**

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formals[L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set x = {4,5,6,7,9}, y = {8,3,7,5,6}

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, l5]

1. Define Data Mining and list any to two basic data mining tasks. Explain the steps involved in KDD Process and Justify the role of data preprocessing in it.

A) Data Mining is the Process of extracting useful patterns and Knowledge from large datasets.

Two Basic Data Mining Tasks:

- Classification.
- Clustering.

2. What is Data Cleaning? Mention two Common data quality issues.

A) Data Cleaning is the Process of identifying and Correcting errors in data.

Two Data Quality issues:

-> Missing Values.

-> Noisy Data.

3. Define Data Integration and list two challenges associated with it.

A) Data Integration is the Process of combining data from multiple Sources into one dataset

Challenges:

- Data Redundancy
- Data Inconsistency

4. What is Covariance Analysis? List two Convariance formula

A:) Convariance measures the relationship between two Variables.

Formulas:

Population Covariance:

$$Cov(X,Y) = \frac{\sum (X - \mu x)(Y - \mu y)}{N}$$

Sample Covariance:

$$Cov(X,Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

5. Suppose two stocks A and B have following values in one week - if stocks are affected by same industry trends, will their prices fall or rise together?

A:) Since both stock prices increase together, the covariance is positive. The stocks rise together and are positively related.

6. Explain Architecture of a Data Mining System.

A:) Components:

- Data Sources.
- Data Warehouse.
- Data Mining Engine.
- Pattern Evaluation
- User Interface.

These work together to discover useful knowledge from data.

7.) Find Covariance

For:

$X = (4, 5, 6, 7, 9)$

$Y = (8, 3, 7, 5, 6)$

A.) Covariance = 0.45

It indicates a negative relationship b/w X and Y.

8.) Describe the major steps in data Pre-Processing. Given a dataset with missing values, outliers. Justify the sequence of steps

A.) Steps:

- Data cleaning.
- Data Integration
- Data Transformation
- Data Reduction
- Data Discretization

9.) A Hospital is analyzing patient age data to identify broad health trends. because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using equal frequency.

A.)

Bin1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin3: 35, 35, 35, 36, 40, 45, 46, 52, 70

10) Explain Data Discretization and its Importance in data mining.

A) Data Discretization

It Converts Continous data into intervals

-> Reduces noise.

-> Simplifies data Analysis.

Equal width: Equal interval size.

Equal frequency: Equal no. of records in each bin.